

Specification of CT Analyzer

1. Scope of work:

The specification covers the design, Manufacturing, Supply , delivery, erection and commissioning of fully automatic multi tap (suitable for at least 3 tap) up to the level of at least 33 KV CT Analyser. Supply and delivery of laptop (with each kit) necessary to run the test kit is also on bidder's scope. All necessary licensed Version of driving software for CT Analyser including product support software, MS Office, antivirus of associated Laptop has to be supplied in CD form separately for each Test Kit/Laptop.

2. PURPOSE:

The CT Analyser is used for measurements of CT-ratio and phase-angle accuracy, polarity, CT winding resistance, CT excitation / saturation (unsaturated and saturated), knee point, demagnetization, Burden impedance, CT residual magnetism. Automatically calculates ratio errors, saturation curves, and knee points. It is microprocessor-controlled variable voltage and current output and precision instrumentation for automatically testing single and multi-ratio CTs of Metering, Protection and PS class, reducing testing time and increasing productivity

3. Qualifying requirement (QR):

- a) The manufacturer must have designed, manufactured, tested and supplied at least Three(3) sets of similar or high end CT Analyser to WBSEDCL / other Power Utilities / other Govt. Departments in earlier occasions within last three financial years. Documents in support of supply of the tendered items to WBSEDCL / other Power Utilities / other Govt. Departments in earlier occasions within last three financial years as mentioned below: .
- a) Purchase Orders, Inspection Offer letter, Despatch Instructions, Signed Challan etc. for completing supply of the item against a particular contract.
- b) Successful Type Test Report carried out at NABL accredited Laboratory in India or equivalent international laboratories. The Type Test should be carried out on similar Type and Model which is going to be offered.
- c) Performance Certificate from other Power Utilities / other Govt. Departments where the Test Kit was supplied earlier within last Three Years.

If the bidder can submit the Credential of the offered set from Power Utility or Government Departments supplied through Channel partners, those certificates will also be acceptable.

In both cases offered set means Sets with same model nos. as those of offered ones.

d) Vendor should be OEM only.

Bidder should have fully equipped technical support office/laboratory having facilities for testing, calibration, diagnosis and repair of equipment in India itself. Bidder should have technical support staff permanently posted in India for technical support after sale. List of Plant and Machinery, tools and tackles to carry out service shall be submitted along with offer.

Statement of orders executed for similar item during last three financial years

4. SPECIFICATION:

Sr. No	WBSEDCL Requirement
1	Equipment should be able to perform following test both manually and automatically.
1.1	Ratio Accuracy :ratio 1 ... 2000 0.02 % :ratio 2000 ... 5000 0.03 % :ratio 5000 ... 10000 0.05 %
1.2	Saturation Test The saturation test can be performed at frequency of 50 Hz up to 2,000 volts as required by IEC regulations it should calculate the rated knee point.
1.3	Phase Angle measurement 3 Digits Range -0 to 360 degree Resolution 1 min Accuracy : +/- 1 min (typical) +/- 6 min(max)
1.4	CT winding resistance measurement (primary and secondary) Accuracy (at 20° C) +/-1%, 0 to 30 Ω ; resolution 1 m Ω
1.5	CT excitation curve The CT Analyzer measures the excitation curve and determines the knee point and other important CT data. Depending on the value of the knee point voltage the CT Analyser identifies the test object as either a Measurement CT or a protection CT.
1.6	polarity check
1.7	Secondary burden measurement: The Burden Test Card allows the measurement of a current transformer's secondary burden by injecting AC current into the load (up to 5 A).
1.8	Knee-point voltage Upto 2 kV can be measured
2	The equipment must have following features:
2.1	Automatic demagnetization of the CT after the test System can automatically demagnetize the CT under test
2.2	Power supply:

	Input voltage 230 V +/- 10% ; 50 Hz; 15A (Max)
2.3	Display: full color, high resolution, LCD
2.4	Environment conditions: Operating temperature: -10 °C ... + 50 °C / 14 °F ... 122 °F
2.5	Automatic assessment according to IEEE C57.13, IEC 60044-1 or IEC 60044-6.
2.6	Remote control Interface:- Full access to all functions of the CT Analyser via a laptop. All necessary licensed version of software shall be provided on free of cost basis. It shall have the facility to export data from the Testing set to laptop through data cable in non-editable format (PDF) customizable reports and storing. The data cable shall be of suitable length.
2.7	Data handling and reporting:- Test reports can be saved/stored on the Compact Flash Card and may be transferred to a PC/Laptop, Data and protocols can be shown on a PC via the Excel TM file loader program, Customizable report templates shall be available for Different standards, classes and applications. Single, multi-core and multi-tap CTs, Three-phase testing, Core testing.
2.8	Weight Volume of products should be less than 0.025m3 and weight less than 20kg otherwise suitable trolley of plastic body with stainless steel shall be supplied by supplier on free of cost basis.
2.9	Safety : As per standard EN60950 and EN61010
3.0	Suitable Lead of 10 mtr length for testing of Current Transformer by CT analyser are to be supplied along with CT analyser.
3.1	The CT Analyser shall have option to be operated both manually as well as through Laptop.

5. Technical Specification of Laptop computer (Should be from known reputed brand)

1) System feature

- a) **Operating System:** Windows 10 professional
- b) Processor: i7, 2.5 GHz, Brand-Intel
- c) Chipset: Mobile HM76 Express,
- d) Cache: 3MB or higher

2) Memory:

- a) System memory: 8GB DDR3 or higher

3) Storage:

Hard drive space: At least 500GB or higher, 7200rpm

4) Display:

- a) Screen Size: 15.6 inch with LED backlit display
- b) Resolution: 1366X768 pixel
- c) Graphics: High definition display

- 5) **Optical Drive:** Super Multi DVD RW with dual layer support
- 6) **Port:**
 - a) 2 X USB 2.0 port, 2X USB 3.0 port
 - b) At least 1 RJ45
 - c) RS 232 port (9 pin serial) / External USB to serial moxa convertor
- 7) **Input Device:**
 - a) Pointer Device: Touchpad
 - b) Keyboard: Full-size keyboard with Numeric Keypad
 - c) Webcam
- 8) **Audio:** integrated Digital Microphone with speaker
- 9) **Network Interface:**
Ethernet, wireless LAN, Bluetooth
- 8) **Dimension & Weight:**
Should be light weight and portable
- 10) **Power supply:**
 - a) Power supply: 230V AC, 50 Hz, 90 W (Not more than) AC Adaptor
 - b) Battery back-up: At least 3 hours
- 11) **Software CD:**
 - a) Product support software
 - b) MS Office Standard
 - c) Reputed brand antivirus software with three years validity.
- 12) **Carry case:** Free with Laptop

6. Defect liability:

- a) All the equipments supplied against this specification shall be guaranteed for a defect liability period of 60 months from the date of receive at DTD store duly passed acceptance testing at DTD lab after delivery.
- b) Any engineering error, omission, wrong provision, equipment failure etc. the equipment shall be attended by the bidder at free of cost during defect liability period.
- c) For any defect in the kit during defect liability period repairing has to be completed within fifteen days. Otherwise substitute set of same type has to be provided within fifteen days from the date of reporting. It will be handed over back to supplier on repair & receipt of original set.
- d) If any defect observed during defect liability period, it is bidder's responsibility to collect kit from site/ consignee to attend the same.
- e) On each failure of test set, supplier shall submit detailed report of failure analysis to Distribution Testing Department, WBSEDCL.
- f) Necessary life time licensed copy of interfacing and analysis software with test kit should be supplied free of cost with test equipment.

7. Demonstration:

During technical scrutiny of the bid, purchaser will ask to demonstrate the offered CT Analyser and requisite Accessories at DTD Laboratory. The bidder shall have to demonstrate the test equipment within 15(Fifteen) days) of intimation from purchaser. It is the responsibility of the bidder to bring all the equipments with necessary software and accessories required to show the performance of the Test Set as per Technical Specification at date & time communicated to them in advance. The bidders are also required to provide the operating manual/Required literatures and send competent Engineers at that. The purchaser will evaluate the suitability of product for their application. The purchaser's decision will be final & unchallenged for acceptance/ non acceptance of product even, if product will comply other tender terms. It is bidder's responsibility to arrange the demonstration with all necessitate at DTD laboratory.

8. Accessories –

Sufficient quantity of Connecting Leads (Properly socket - terminated at Both End) for all voltage, current and Binary Input/Outputs each of length at least 2 Meter long,, Input Power Cable and other relevant accessories required for “CT Analyser” and the “Laptop with Printer”. The laptop shall be of reputed make i.e. HCL, HP, Toshiba, Lenevo. The printer shall be of Laser Jet type.

9. Literature & Technical manual - The manufacturer is to provide at least two sets of Literature & technical manual per set.**10. Type Test:-**

The test set shall be Type Tested for EMI, EMC & Degree of Protection for IP 54 conducted from third party NABL accredited laboratory or Government Departments within last 05(Five) years.

11. COMMISSIONING:-

The supplier shall have to commission the equipment at consignee's premises as Desired by the consignee. Necessary standard tests shall be done at the consignees premises during the commissioning of the equipment. The consignee's staff shall be trained by the manufacturer's Engineer for Satisfactory operation, maintenance and trouble shooting upto their full satisfaction.

---END---

GARAUNTEED TECHNICAL PARTICULARS

Sl. No.	Item to be replied	To be filled up by the Bidder
1	Name of the Manufacturer	
2.	Office Address	
3.	Manufacturing Unit Address	
4.	Whether the bidder is the OEM of the tendered items (Yes/No)	
5.	Whether the bidder has comprehensive service centre along with competent Engineers, if yes, mention the address.	
6.	Model Names of the CT Analyser and their Accessories including that of Laptop offered by them(Mentioning Item wise Model No.)	
	a)CT Analyser Set	
	b) Laptop computer	
	c)Printer	
7.	Whether the CT analyser comply all the criterion of the Technical Specification	
8.	Whether data from the CT analyser can be brought to the Laptop from the CT analyser through data cable.	
9.	Length of data cable	
10.	Whether agreeable to demonstrate the equipment at the fixed date in WBSEDCL cable network	
11.	Whether data can be brought from the CT Analyser set in the form of non editable (PDF) report format	

12.	Size and weight of the CT Analyser Test Set?	
13.	Length of the cable(Lead) and it's current rating for sending the requisite current through CT during Testing.	
14.	Whether the Lead for Testing is provided with copper Clamp with sufficient tightness and current carrying capacity.	
15.	Whether the CT analyser can be operated from the manually as well as from the Laptop?	
16.	Whether agreeable to train WBSEDCL Engineers at their full satisfaction about operation of entire range of instrument and for physical demonstration of the equipment after placement of P.O, if qualified.	
17.	Whether agreeable to guarantee each and every components of the instruments for 5 years.	
18.	Whether submitted satisfactory performance of each and every components of the equipment from any power utility/Govt. department for a period of 3 years.	
19.	Whether Type Test report for EMI, EMC & Degree of Protection for IP 54 conducted from third party NABL accredited or equivalent international laboratory has been submitted? If yes mention the Type Tests conducted.	

Signature of Company representative with seal